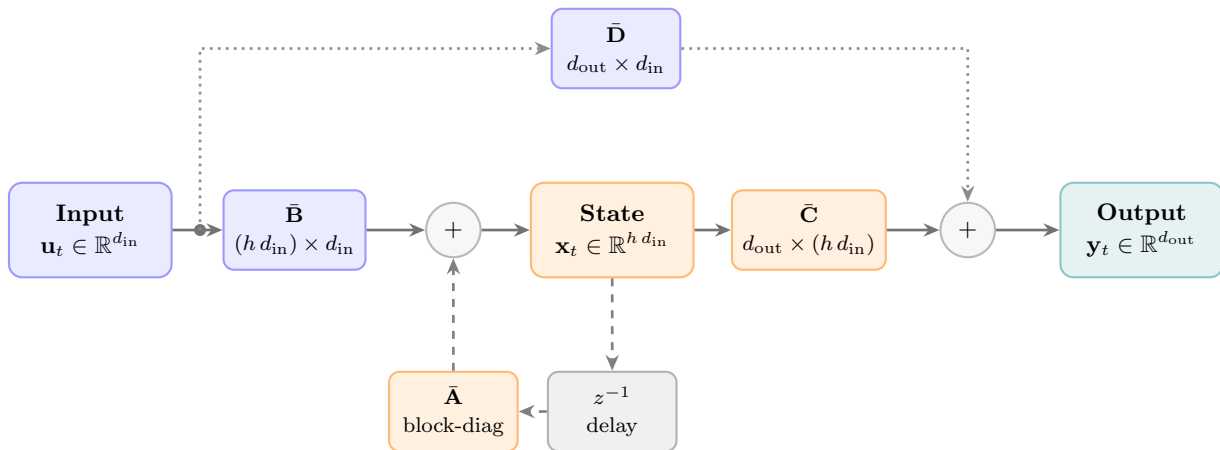


Discrete-time MIMO structured state space model

feedforward path (skip; often $\bar{\mathbf{D}} = 0$)



feedback path (hidden-state recursion through the delay register z^{-1})

$$\text{state: } \mathbf{x}_t = \bar{\mathbf{A}} \mathbf{x}_{t-1} + \bar{\mathbf{B}} \mathbf{u}_t$$

$$\text{output: } \mathbf{y}_t = \bar{\mathbf{C}} \mathbf{x}_t + \bar{\mathbf{D}} \mathbf{u}_t$$